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Tectonophysics 363 (2003) 103–125
TECTONOPHYSICS
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Differences in paleomagnetic interpretations due to the choice of statistical, demagnetization and correction techniques: Kapuskasing Structural Zone, northern Ontario, Canada

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Received 7 May 2002; accepted 22 November 2002

Abstract

The Kapuskasing Structural Zone (KSZ) reveals a section through the Archean lower crustal granoblastic gneisses. Our new paleomagnetic data largely agree with previous work but we show that interpretations vary according to the choices of statistical, demagnetization and field-correction techniques. First, where the orientation distribution of characteristic remanence directions on the sphere is not symmetrically circular, the commonly used statistical model is invalid [Fisher, R.A., Proc. R. Soc. A217 (1953) 295]. Any tendency to form an elliptical distribution indicates that the sample is drawn from a Bingham-type population [Bingham, C., 1964. Distributions on the sphere and on the projective plane. PhD thesis, Yale University]. Fisher and Bingham statistics produce different confidence estimates from the same data and the traditionally defined mean vector may differ from the maximum eigenvector of an orthorhombic Bingham distribution. It seems prudent to apply both models wherever a non-Fisher population is suspected and that may be appropriate in any tectonized rocks. Non-Fisher populations require larger sample sizes so that focussing on individual sites may not be the most effective policy in tectonized rocks. More dispersed sampling across tectonic structures may be more productive. Second, from the same specimens, mean vectors isolated by thermal and alternating field (AF) demagnetization differ. Which treatment gives more meaningful results is difficult to decipher, especially in metamorphic rocks where the history of the magnetic minerals is not easily related to the ages of tectonic and petrological events. In this study, thermal demagnetization gave lower inclinations for paleomagnetic vectors and thus more distant paleopoles. Third, of more parochial significance, tilt corrections may be unnecessary in the KSZ because magnetic fabrics and thrust ramp are constant in orientation to the depth at which they level off, at approximately 15-km depth. With Archean geothermal gradients, primary remanences were blocked after the foliation was tilted to rise on the thrust ramp. Therefore, the rocks were probably magnetized in their present orientation; tilting largely or entirely predates magnetization.
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Keywords: Paleomagnetism and deformation; Orientation statistics; Kapuskasing Structural Zone; Canadian Shield

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doi:10.1016/S0040-1951(02)00667-4

1. Introduction

The Kapuskasing Structural Zone (KSZ) of northern Ontario is a well-documented example of upthrust lower crust, for which most contemporary research is documented by Percival (1994) and Percival and West (1994) (Fig. 1). The zone thrust up Archean lower crustal granoblastic gneisses, dated at 2750–2500 Ma, that rose on a ramp from a subhorizontal decollement, now at a depth of ~ 15 km (Geis et al., 1990; Percival and Card, 1983). Intracratonic uplift and cooling occurred at ~ 1900 Ma, exposing a dextrally sheared lower-crust section >15 km thick. KSZ shows strong positive gravity and magnetic anomalies which, together with recent seismic surveying, indicate that lower crustal rocks were thrust up from the west over the Archean Abitibi Greenstones, which crop out to the east of the Kapuskasing Structural Zone (Atekwana et al., 1994).

On the east, KSZ is bounded by the Ivanhoe Lake Fault Zone. Interpretations of seismic profiles suggest that it dips ~ 45° to the northwest, which is steeper than the thrust (Fig. 1b). The seismic profile documents reflectors to a depth of approximately 15 km, emphasizing contributions from gently inclined reflectors. They appear to show that the lower crustal slice maintains constant orientations of its internal fabrics and of its internal seismic reflectors to depths ≥ 10 km. The thrust surface and seismic reflectors within the lower crustal granulites are subparallel to the magnetic foliations (Borradaile et al., 1999) and dip ~ 20°, more gently than the Ivanhoe Lake Fault (Shaw and Percival, 1990; Leclair et al., 1994). Pressure estimates from petrogeneses and radiometric ages constrain the pre-thrusting uplift rate to approximately 0.03 km/Ma following late Archean tectonism at 2670 Ma (Percival, 1994; Percival and West, 1994). This is quite modest by comparison with modern tectonic environments and is similar to contemporary continental denudation rates. A wealth of recent geochronological, geobarometric and geothermal data permit some reasonable constraints on uplift rates (Hanes et al., 1994; Krogh and Moser, 1994; Mäder et al., 1994; Percival and Peterman, 1994; Percival et al., 1994; Williams et al., 1992). During thrusting of the Kapuskasing granulites to the surface at ~ 1900 Ma, the uplift rate was four times higher, ~ 0.12 km/Ma, but still less than Phanerozoic orogenic uplift rates. Conservative geothermal gradients were previously used (30 °C/km), although values closer to 50 °C/km may be possible based on our knowledge of metamorphic studies. Such imprecision makes it difficult to estimate the depth of the Curie isotherm or any blocking temperature isotherm with sufficient precision to fix the age of any paleomagnetic signal. Comparison of paleopole locations with the tentative apparent polar wander paths (APWPs) is a more optimistic strategy.

The granoblastic rocks of the Kapuskasing Structural Zone (KSZ) lack anisotropic features that can be used in structural mapping to determine finite strain directions and infer kinematics. However, magnetic fabrics reveal subtle tectonic microfabrics precisely and reproducibly (Borradaile, 1988; Borradaile and Henry, 1997; Hrouda, 1982). Recently, we applied the techniques of measuring anisotropy of magnetic susceptibility (AMS) and anisotropy of anhysteretic remanence (AARM) in the KSZ (Borradaile et al., 1999). The AMS method determines the anisotropy of mean susceptibility tensor for all grains, but primarily that of the aligned paramagnetic mafic crystals that dominate >95% of the induced magnetic response to a low applied field. This low-field technique was applied to 263 specimens before any demagnetization experiments that could disturb the AMS. It is known that high-field exposure may change magnetite's AMS (Potter and Stephenson, 1990), and that oxidation during thermal demagnetization may change AMS even more significantly by the alteration or creation of oxides (Borradaile and Lagroix, 2000). Our AMS results mostly define the orientation distributions of the silicate matrix. AARM isolated the orientation distribution for the assemblage of remanence-bearing grains, a technique explained by Jackson (1991). Our previous work on magnetic mineralogy, hysteresis and Curie temperatures confirms that the remanence-bearing phase is almost exclusively magnetite, further confirmed by the upper limit of the blocking-temperature spectra. Borradaile et al. (1999) confirmed this from Curie-temperature determinations, although minor remanence components were attributed to localized hematite alteration, and a mineral with a Curie temperature ~ 260 °C, which may be a titaniferous iron oxide (Dunlop and Özdemir, 1997; McEnroe et al., 2001). AARM defines the principal directions for the orientation distribution of magnetite's grain shapes. The technique is more restrictive on the suitability of

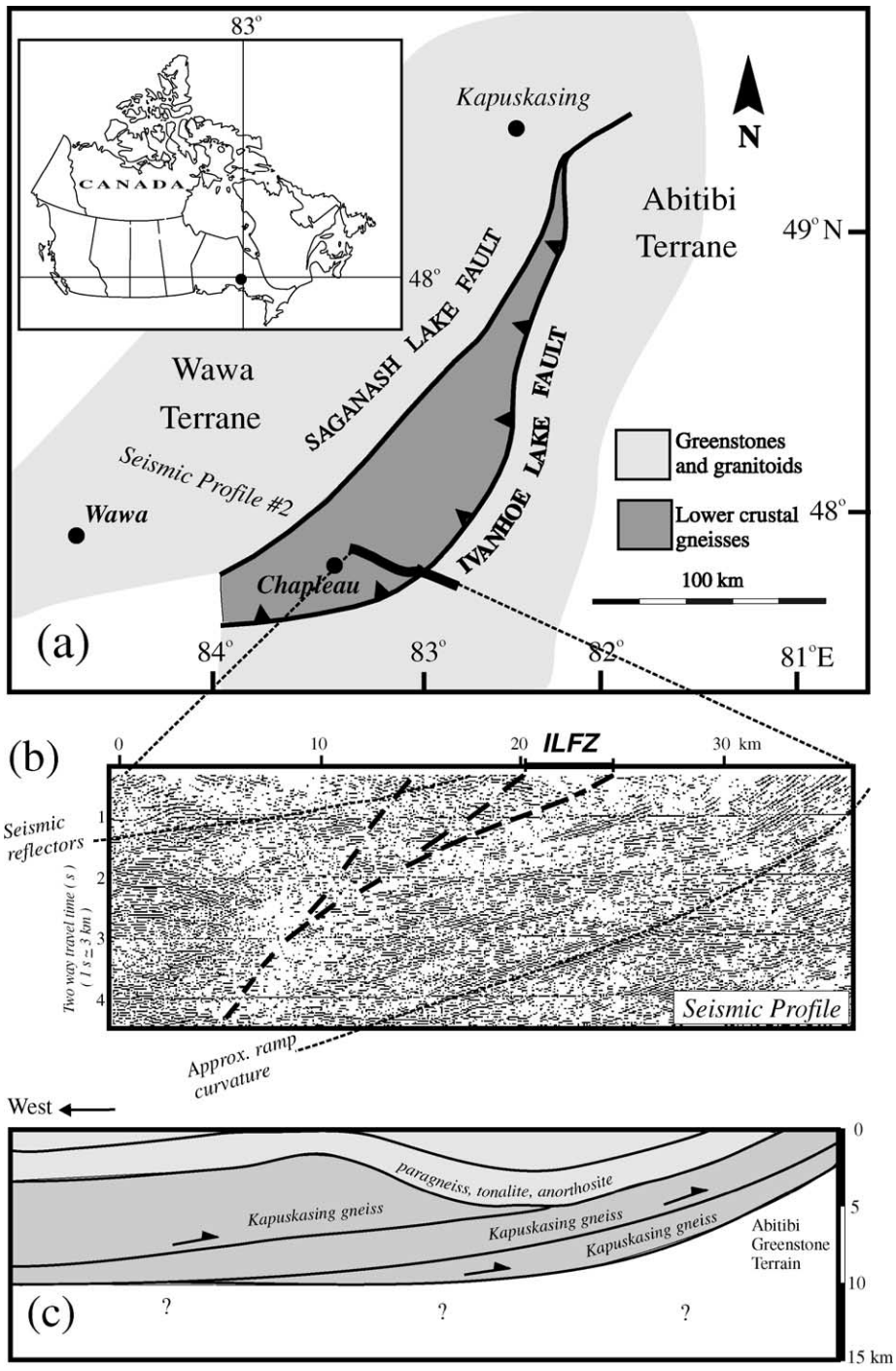


Fig. 1. Location of the Kapuskasing Structural Zone (KSZ) in the Superior Province of the Archean Shield of Northern Ontario. Below: principal seismic reflectors in an ENE–WNW traverse after Hanes et al. (1994), Percival (1994), Percival and West (1994), Percival et al. (1989) and Shaw and Percival (1990). Note that the apparent dip of the seismic reflectors is slightly less than the true dip in this section. At the surface, the reflectors are subparallel to magnetic foliations (e.g. Figs. 2 and 6).

the magnetic mineralogy, but was successful for 156 of the specimens. Of course, the cores used for AARM fabric studies, having been exposed to high fields, are not suitable for subsequent paleomagnetic work. Therefore, for paleomagnetic work, new cores were prepared from the same large oriented hand specimens that provided the AARM data.

The summarized magnetic fabric results from the KSZ correspond to major tectonic features. The orientations of principal axes of the magnetic fabric tensors are shown by two different methods. First, the traditional contoured density plots of minimum and maximum axes are shown for AMS (Fig. 2a and b) and AARM (Fig. 2d and e). The densities of the orientation distribution disregard the mean susceptibilities of the specimens; all specimens are weighted equally. This means that the influence of magnetite-rich specimens is subdued in AMS and the AMS density distribution mainly reflects the orientation distribution of the silicate matrix. For AARM, of course, the density

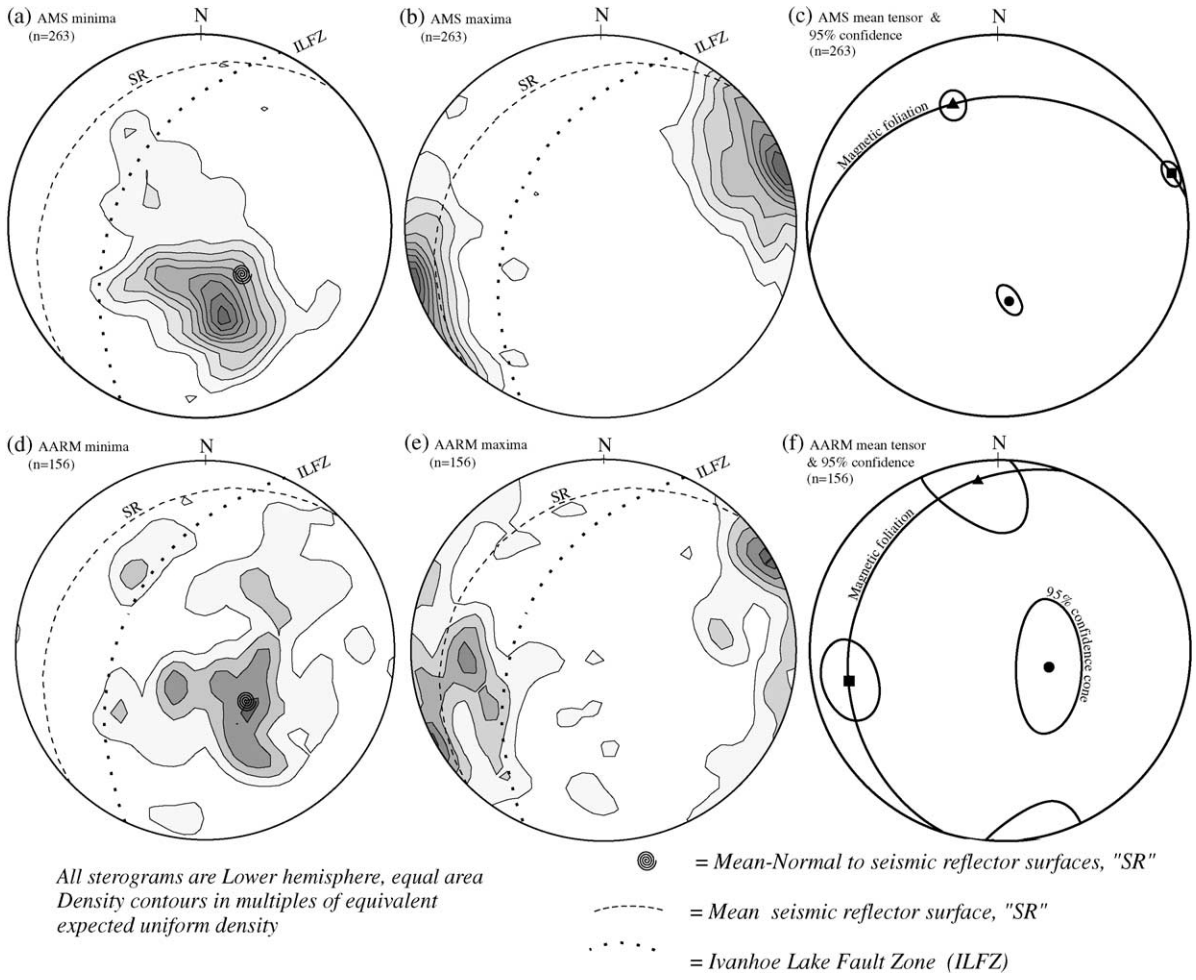


Fig. 2. Orientation distributions of magnetic anisotropy fabrics; AMS = anisotropy of low field magnetic susceptibility; AARM = anisotropy of anhysteretic remanent anisotropy; ILFZ = Ivanhoe Lake Fault Zone. (a) The normal to seismic reflectors is similar in orientation to the AMS minimum which is normal to AMS foliation. (b) AMS maximum, defining cryptic mineral lineation, lies in the plane of the seismic reflectors. (c) Mean tensor for AMS, not standardized, differs slightly from density-contour distribution in (a) and (b). (d) Normal to AARM foliation corresponds to the normal of seismic reflectors. (e) AARM lineation is subhorizontal, in plane of the seismic reflectors. (f) Mean tensor for AARM, not standardized, corresponds well with density distributions of (d) and (e).

plots isolate the orientation distribution of magnetite. The use of the mean-tensor plot requires some consideration (Borradaile, 2001). In Fig. 2c and f, the mean tensors for AMS and for AARM are shown for specimens, without standardization. Standardization or normalization is a procedure that weighs the orientations of individual specimens according to their mean susceptibility. Consequently, the mean tensor axial orientations and their 95% confidence cones are influenced more by high-susceptibility specimens. In the case of AARM, the mean-tensor orientations correspond well with the density plots of the maximum and minimum principal AARM axes (cf. Fig. 2f, d and e). This is because only the magnetite subfabric contributes to AARM; that subfabric is reasonably homogeneous and each specimen has approximately similar anhysteretic susceptibility. In contrast, the AMS mean tensor (Fig. 2c) does not correspond quite as closely to density plots of its principal axes (Fig. 2a and b). This is due to the contribution of a high-susceptibility, oblique magnetite subfabric that in some specimens deflects the net AMS fabric away from the orientation of the silicate matrix.

Petrographic and structural observations show that the magnetite subfabric is younger than the silicate fabric and that the two fabrics developed successively in a noncoaxial strain history. The silicates grew first, defining a very feeble metamorphic foliation due to the low shape anisotropy of grains in the granulite facies. Subsequently, magnetite developed in part by exsolution from mafic silicates. Magnetic anisotropy isolates the orientation of the plane of flattening at a different part of the kinematic history; AMS detects the orientation distribution of the silicates but it with contributions from the accessory oxides. AARM isolates the younger orientation distribution of magnetite. This has been recognized along transpressed Archean terrain boundaries in adjacent regions (Borradaile and Dehls, 1993; Werner and Borradaile, 1996; Borradaile et al., 1999). AMS maxima define the mean extension axis which corresponds to the minimum compressive stress during crystallization of the silicate fabric (Fig. 2b and c). AARM maxima indicate mean extension during crystallization of the magnetite subfabric (Fig. 2e and f). The AMS and AARM mean foliations are separated by a dihedral angle $<15^\circ$ (Fig. 2). The silicate fabric mainly responded to EW extension on a north-dipping foliation whereas the subsequent magnetite fabric developed during W-plunging extension on a WNW-dipping foliation (cf. Fig. 2c and f).

It is important to know how deep the fabrics persist without change in orientation because this will influence the decision concerning a paleomagnetic tilt correction. In this region, estimates of the required tilt corrections have decreased with the advancement of structural and seismic knowledge from 30° to 15° (Constanzo-Alvarez and Dunlop, 1988) to $<10^\circ$ (Symons et al., 1994). Fortunately, the Lithoprobe project provided an ENE–WNW seismic profile (Fig. 1b) which revealed diffuse pervasive seismic reflectors throughout the KSZ granulites to ≥ 15-km depth. Their true dip is unknown as they have only been recorded in the profile plane, and the contribution of shallow reflectors is recorded at the expense of others. Nevertheless, in the profile, their mean orientation is parallel to that of the average AMS and AARM foliations projected from the surface. Thus, we propose that seismic reflectors are an expression of the metamorphic fabric defined by AMS and AARM. The orientation of all three fabrics may therefore be similar to depths ≥ 10 km, following the orientation of the seismic reflectors (≥ 15 km), and at a similar or slightly lower inclination than the thrust ramp (Percival and West, 1994). Thus, the KSZ ramp geometry may have been exhumed without significant tilting. Blocking-temperature isotherms ($\sim 450-550$ °C) would have been achieved at original depths ≥ 10 km using conservative estimates of Archean geothermal metamorphic gradients (~ 2600 Ma). This would mean that the granulites were magnetized in their present orientation, and any significant tilt correction may be unnecessary. However, Constanzo-Alvarez and Dunlop (1988) recognized a spatial distribution of unblocking temperatures and a combination of heterogeneous tilt corrections across the strike of the KSZ that permitted a model involving late tilting of preserved A-component remanences.

2. Choices in the treatment of paleomagnetic data

Paleomagnetism was studied extensively in the KSZ, accompanying the Canadian Lithoprobe program of seismic reflection studies. Results were chiefly obtained by Bates and Halls (1990, 1991a,b), Buchan et al. (1993), Constanzo-Alvarez and Dunlop (1988),

Constanzo-Alvarez et al. (1993), Ernst and Buchan (1993), Ernst and Halls (1984), Halls et al. (1994), Vandall and Symons (1990) and Symons and Vandall (1990). Symons et al. (1994) summarized the findings in a comprehensive synthesis. Our resampling increases the existing database by approximately 8%, which is sufficient to validate our results by comparison with others. Since the sample size is adequate and the main results are corroborated by earlier work, it permitted us to experiment with different paleomagnetic procedures. Our contribution is not to bring different data but to show that different procedures may lead to different interpretations of KSZ tectonics and of the global movements of the KSZ rocks by comparison with the tentative apparent polar wander paths (APWPs) for the late Archean and with the more confident paths for the Early Proterozoic rocks (Buchan et al., 1990, 1993, 1996, 2000, 2001; Dunlop, 1983; Constanzo-Alvarez and Dunlop, 1988, Constanzo-Alvarez et al., 1993; Irving and McGlynn, 1981; Symons and Vandall, 1990; Symons et al., 1994; Vandall and Symons, 1990).

The variations in procedure with which we have experimented involve the following items. First, the results of alternating field (AF) and thermal demagnetization define slightly different paleopole locations. Second, characteristic remanence vectors (ChRM) yield slightly different confidence cones and central orientations if the sample is considered to have been drawn from a Bingham population (Bingham, 1964), rather than from the commonly assumed Fisher population (Fisher, 1953). Third, a tilt correction may be unnecessary due to the apparent correspondence of uniformly dipping metamorphic and magnetic foliations in the KSZ with similarly oriented seismic reflectors which persist at constant orientation to depths ≥ 10 km.

Our results are compared against the backdrop provided by the apparent polar wander paths (APWP) available from the literature. It is appreciated that the APWPs are tentative in nature but we caution the reader that they are probably just as reliable as APWPs from some tectonized Cenozoic sequences. This is because magnetization mostly postdates strain during post-tectonic exhumation. Thus, contentious corrections for tilt, folding and strain are unnecessary. It is acknowledged that the precision of ages is generally poor, with the exception of recent findings by Buchan et al. (1993, 1996, 2000, 2001) and some of the earlier data tabulated by Vandall and Symons (1990). The APWPs are based on the results of the above cited workers plus the results of Dunlop (1979, 1983, 1984a,b, 1985) from adjacent areas and are summarized in Fig. 7a. The reader is cautioned to note the stretching of longitude at high latitudes in Gall's World map projection.

Previous paleomagnetic studies included 235 sites and 1652 specimens reported from all preceding publications available to us. This almost certainly implies some site duplication because the area is poorly exposed. The interpretation of these data by Symons et al. (1994) reveals a consensus of opinion, with the exception of minor debates concerning polarity switches, the amount of tilting and of vertical-axis rotation due to local tectonics. The principal conclusions obtained almost exclusively by thermal demagnetization appear to be as follows. The oldest geologically significant signal reported is a normal A-component magnetization dated at ~ 2550 Ma (37% of data). The youngest significant paleomagnetic event is a reversed B-overprint (77% of data), timed variously between approximately 1100 and 1800 Ma. The latter is attributed to remagnetization during Proterozoic exhumation of the KSZ.

For our study, alternating field (AF) demagnetization was performed in a Sapphire Instruments demagnetizer with triple-layer, mumetal shielding (<1 nT residual field) and peak AF between 80 and 200 mT, depending on the coercivity in question, incremented in steps as small as 0.1 mT. At least 20 increments of AF demagnetization were used per specimen. Internal circuitry corrects for spurious ARM and our software automatically alerts the user to spurious gyroremanence contributions (Stephenson, 1983). Thermal demagnetization was performed in a triple-layer, shielded, non-inductively wound, ceramic furnace with a residual field <1 nT. This apparatus was manufactured in our laboratory. A few later demagnetization experiments used a recently acquired MMTD80 thermal demagnetizer. At least 10 steps of demagnetization were performed on each specimen from 80 to >600 °C. Rotatable, three-dimensional vector plots were viewed interactively during demagnetisation, as were conventional, two-dimensional vector graphs (Zijderveld, 1967). This permitted simultaneous determination of the direction of significant vector components by a

Table 1
A-components for the Kapuskasing Structural Zone

A-components	Sites (N)	ChRM (n)	Vector mean (Dec/Inc)	Fisher k	Fisher α_{95}	T_{UB} (°C)	Hc (mT)	Age (Ga)
Constanzo-Alvarez and Dunlop (1988)	8	46	023/+53	49	7	>500		2.55
(Shawmere traverse)	5	23	050/+62	138	5	>400	15–100	2.55
Symons and Vandall (1990)	15	76	023/+50	52	11	350–585	High	
Our data (Normal ChRMs)	41	83	see Table 4			20–580	5–200	

ChRM = number of vectors; for vector mean; Dec = declination; Inc = inclination.
α_{95} = Fisher (1953) semi-angle of 95% confidence cone about direction, in degree; k = concentration parameter.
T_{UB} = typical unblocking temperatures. Hc = typical coercivities. Vector mean involves no structural correction.

principal component analysis subroutine (Kirschvink, 1980). Six or more data points were used to define linear vector components for each stable remanence component using interactive Windows-style software for the Molspin and JR5a spinner magnetometers written by Borradaile.

In our specimens, raw NRM vectors were mostly downward-seeking and broadly dispersed around the present geomagnetic-field direction. The specimens had been stored in stable orientations for more than a year before the initial measurements. Prior to demagnetisation, the mean NRM intensity was 200 mA/m. The potential for magnetisation is almost twice as high because the mean value of the seven anhysteretic remanences of each of the 156 specimens measured during AARM analysis was 400 mA/m, using a bias DC field of 0.1 mT, over an AF window from 60 mT down to zero. This gives a mean theoretical Koningsberger ratio of $Q_t \approx 2$ (Dunlop and Özdemir, 1997), which could be considered as an estimate of the maximum potential value of Q in nature. The conventional Koningsberger ratio has a mean value of $Q \approx 0.33$, comparing NRM intensities with the induced magnetization at 50 nT, using actual average bulk low field susceptibilities of the specimens (the average bulk susceptibility for the suite of specimens was $\bar{k} \sim 7500$ μSI).

Our independent survey of 41 sites and 114 specimens revealed that 78% of the specimens carry normal characteristic remanences (ChRM) with reversed secondary components. The remaining 22% of specimens carried only reversed components. There is no geographic control on the distribution of polarity, declination or inclination of remanence components among our sites. ChRMs isolated by principal component analysis are listed in Tables 1 and 2, shown in the stereograms of Figs. 4 and 5 and summarized in the stereogram of Fig. 6. Thermal demagnetization successfully isolated 31% of the ChRMs, of which 73% were normal and 27% were reversed. AF demagnetization successfully isolated 69% of the ChRMs, of which 77% were normal and 23% were reversed. Normal remanences (Table 3) were identified almost twice as often in our study, as compared to previous

Table 2
B-components for the Kapuskasing Structural Zone

B-components	Sites (N)	ChRM (n)	Vector mean (Dec/Inc)	Fisher k	Fisher α_{95}	T_{UB} (°C)	Hc (mT)	Age (Ga)
Constanzo-Alvarez and Dunlop (1988) Shawmere	4	16	115/−68	17	17	High/Bimodal		1.1
Constanzo-Alvarez et al. (1993) Shawmere	18	72	097/−68	12	10	High/Bimodal		1.86 E 1.1 W
Symons and Vandall (1990)	18	173	073/−68	52	5	580–675	0–100	1.86
Our data (Reversed ChRMs)	14	31	see Table 4			450–580	0–180	?

ChRM = number of vectors; for vector mean; Dec = declination; Inc = inclination.
α_{95} = Fisher (1953) semi-angle of 95% confidence cone about direction, in degrees.
T_{UB} = typical unblocking temperatures. Hc = typical coercivities.
Vector mean involves no structural correction.

Table 3
Coercivity ranges of normal and reverse characteristic remanences

	ChRM (n)	H_{max} (mT)	H_{min} (mT)	PCA (n)	ASD
Normal	95	72.56	7.56	12	8
Reversed	18	87.21	19.95	10	12

ChRM = number of specimens yielding meaningful characteristic remanences.
H_{MAX} = mean of upper coercivity limit of ChRM.
H_{MIN} = mean of lower coercivity limit of ChRM.
PCA = mean of number of points used to fix vector by principal component analysis (PCA).
ASD = mean of angular standard deviations for PCA vectors.

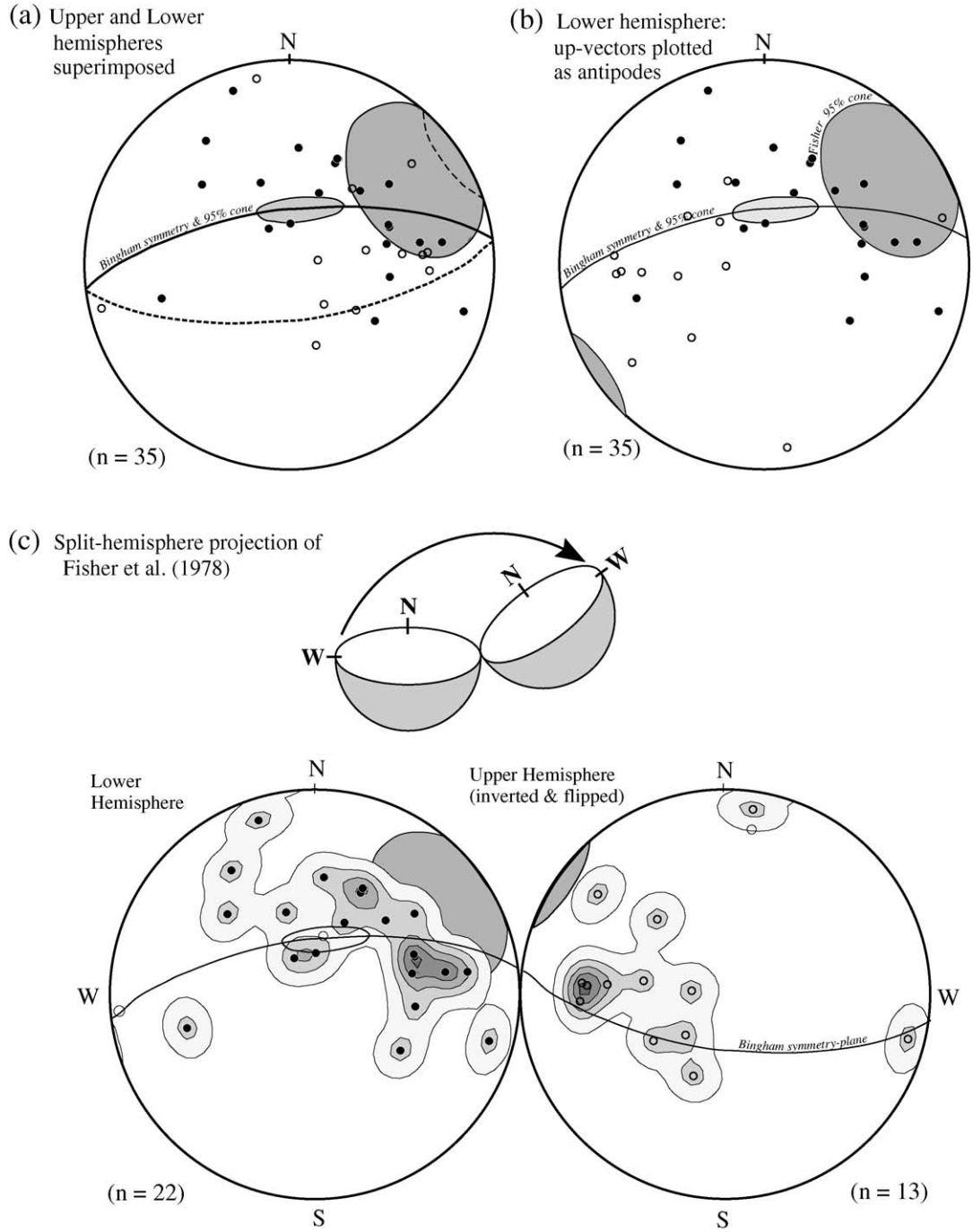
studies. This may be due to our more extensive use of AF demagnetization, which was more successful in isolating the normal (A) component.

By tradition in paleomagnetism, the representation of distributions with both up and down vectors may use the combined projection of the upper and lower hemispheres simultaneously, differentiating points on the upper hemisphere with open symbols, from those on the lower hemisphere with solid symbols (Fig. 3a). Alternatively, for small numbers of points and ease of comparison with structural orientations, all data may be plotted on the lower hemisphere, representing up-vectors by their antipodal projection onto the lower hemisphere (Fig. 3b). Either of these traditional approaches may present difficulties in visualization where there are large numbers of data or where the paleomagnetic vectors must be compared with large numbers of structural axes (e.g. normals to foliation, mineral lineation or magnetic fabric axes as in Fig. 2). Like structural axes, large numbers of vectors benefit from the evaluation using density contours that objectively characterise an orientation distribution. Density contours can be useful in the evaluation of paleomagnetic data but it may be that presentation difficulties have limited their use since their introduction in this context (e.g. Stupavsky and Symons, 1982). This is unfortunate; density contours are particularly valuable in data reconnaissance since they are independent of any preconceived statistical model, unlike confidence regions around mean directions. Density contours may be especially useful in paleomagnetic studies of deformed rocks in which remanence vectors may be dispersed due to the effects of magnetic anisotropy, stress-remagnetization or finite strain effects (Borradaile, 1997; Werner and Borradaile, 1996).

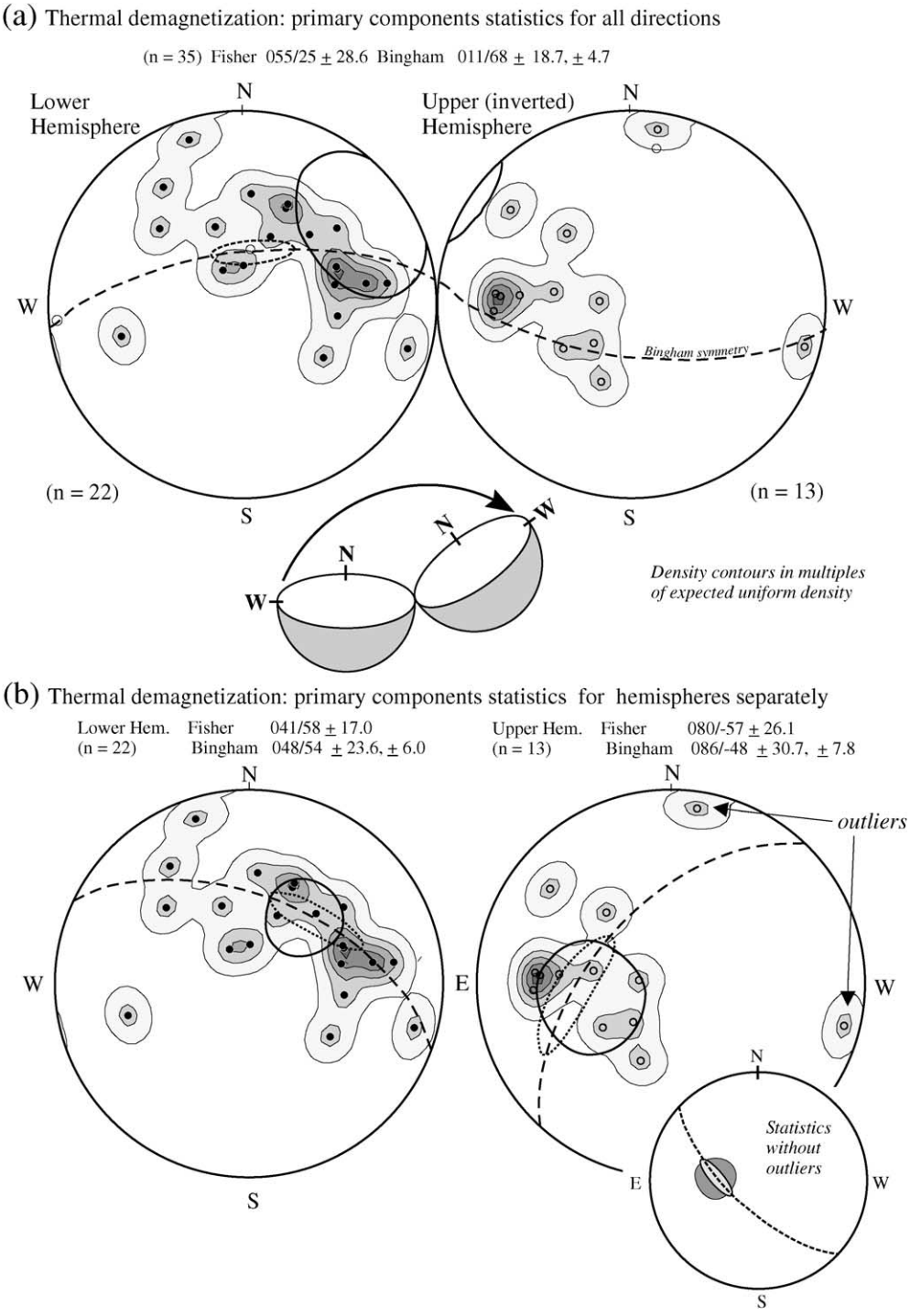
Fisher et al. (1987) introduced a split-stereogram projection that permits inspection of vector orientations and their density distributions on both upper and lower hemispheres, simultaneously. The downward-seeking vectors are plotted with their density contours on the lower hemisphere in a conventional manner. On an adjacent circular diagram, the upper hemisphere is plotted. Our use of the split stereogram is slightly different from Fisher et al. (1987) and is shown in Fig. 3c (inset). Whereas Fisher et al. unhinge the hemispheres about the north mark, we unhinged about the east mark, flipping the upper hemisphere up-and-over eastwards. This may be visually more intuitive to geologists (Fig. 4a, inset) as north is retained at the top of each diagram in our usage. Structured patterns in density contours may be readily traced from upper to lower hemisphere using the split projection. It is also sometimes easier to identify clusters and outliers from those concentrations (Fig. 4b), although in the other orientation distributions of vectors, we were unable to justify exclusion of any stereogram points.

The diagram is used to illustrate thermally isolated components (Fig. 4a) and AF-isolated components (Fig. 5), from our Kapuskasing data, and summarizes all paleomagnetic, microfabric data and selected structural data in Fig. 6. This projection also makes it is easier to verify unimodality and whether or not the mode is symmetrically circular, so that the appropriate statistical model may be selected.

Fig. 3. Orientation distributions of vectors that include upwards- and downwards-directed vectors may be represented in different ways on stereograms. Here, three techniques are shown for the same data in this study, the thermally isolated primary characteristic A-components. (a) Up and down vectors may be simultaneously shown on an upper and lower hemisphere that are superimposed in plane projection. (b) A lower hemisphere projection may be used, plotting upwards-directed vectors antipodally on the lower hemisphere. (c) The split and flipped spherical projection has the advantage that density contours may be readily used and clusters may be identified on both hemispheres. Another advantage is that clusters, and outliers from clusters, may be readily identified. In practice, it may be advisable to centre the orientation distribution first, about some significant axis (e.g. the mean direction or some structural axis) so that the sphere is split on some plane other than the horizontal one. As shown here, the split hemisphere projection is slightly modified after Fisher et al. (1987).



Three styles of projection for the same data, A-components derived by thermal demagnetization: (n = 35) Fisher mean 055/25 (± 28.6) Bingham 011/68 (± 18.7, ± 4.7)


(a) Thermal demagnetization: primary components statistics for all directions
(n = 35) Fisher 055/25 ± 28.6 Bingham 011/68 ± 18.7, ± 4.7
Lower Hemisphere
Upper (inverted) Hemisphere
N
W
S
E
Bingham symmetry
(n = 22)
(n = 13)
Density contours in multiples of expected uniform density
(b) Thermal demagnetization: primary components statistics for hemispheres separately
Lower Hem. Fisher 041/58 ± 17.0
(n = 22) Bingham 048/54 ± 23.6, ± 6.0
Upper Hem. Fisher 080/-57 ± 26.1
(n = 13) Bingham 086/-48 ± 30.7, ± 7.8
outliers
Statistics without outliers

By tradition, and with good reason, most samples of paleomagnetic vectors are considered to be drawn from a population of points on the sphere that has circular symmetry about the mean vector. The mean vector is determined without assuming any statistical model for the population by vector summation. However, for paleomagnetic purposes including determination of paleopole locations and APWPs, it is essential to establish the confidence limits about the mean vector, usually at the 95% level. Formal hypothesis testing is rarely pursued, most workers using visual comparison of confidence regions to ascertain similarity or difference between mean vectors. However, hypothesis testing of orientation distributions is discussed by Fisher et al. (1987).

The assumption that the population has circular symmetry about the mean vector is central to the statistical treatment introduced by Fisher (1953) and is almost universally used and almost universally appropriate in paleomagnetism. This simple model has sound theoretical and experiential support in the case of stable paleomagnetic vectors. First, when an isotropic material is instantaneously magnetized, a dipole geomagnetic field can only produce an axial–circular–symmetric distribution of remanence vectors. Second, the anisotropy of most rocks is so low that remanence dispersal due to the internal fabric is rare (Fuller, 1963, 1970; Hrouda, 1982; Rochette et al., 1992; Stephenson et al., 1986). Since orientation distributions have a finite angular range, variance is a difficult concept of restricted use. However, the inverse concept of a precision parameter (κ) for n orientations is readily approximated by a concentration parameter $k=(n-1)/(n-R)$, where R is the resultant (e.g. Butler, 1992). For the case of a symmetrically circular distribution of vector endpoints on the sphere, the Fisher model permits the easy calculation of the 95% confidence limit about the vector mean by the approximation $\alpha_{95}=140°/(kn)$. This expression approximately defines α_{95} in degrees as the apical angle of a confidence cone that is circular in cross-section and coaxial with the mean vector. Its projection on an equal-area stereogram is non-circular, except when the mean vector is precisely vertical. In spite of this, the equal-area stereogram is obviously retained because the areal distortion of the equal-angle stereogram causes far greater visual bias in interpreting orientation distributions. In this paper, all projections are equal in area, and most are in the lower hemisphere except where designated upper hemispheres appear on the right-hand side of the hinged-net device (Figs. 4–6).

The circular distribution of vector termini on the sphere is a good model in most paleomagnetic studies. However, increasingly more paleomagnetic work is being performed on tectonized rocks. Whether the remanences were acquired before or during tectonic deformation, circular distributions of vectors on the stereogram are less likely. The most common alternative vector distribution lies along some part of a great circle forming an ellipse. A statistical model for their elliptical distribution on the sphere was established by Bingham (1964) with practical applications rendered easier by Mardia and Zemroch (1977). Many other distributions can be investigated (Fisher et al., 1987; Mardia and Jupp, 2000) and although small circle distributions may arise during folding of older remanences, there are good geological and theoretical reasons for expecting almost all single event, homogeneous and macroscopic structural processes to produce great circle dispersions. Most paleomagnetic vector distributions follow either the Fisher distribution or the Bingham distribution, which give rise to the circular–symmetrical and the elliptical–orthorhombic patterns, respectively. An elliptical distribution on the sphere may be modelled with simpler mathematics than the Bingham model using the bivariate normal distribution on the sphere (Le Goff et al., 1992). The more simply calculated confidence

Fig. 4. Primary A-components from thermal demagnetization isolated by principal component analysis (Kirschvink, 1980). There were at least 12 steps of thermal demagnetization per specimen. The angular distribution of vector may not, as here, conform to an idealized, single symmetrically circular cluster. Thus, treatment by Fisher (1953) statistics may be less appropriate than by statistics based on the more general model of Bingham (1964). Fisher confidence cone is shown centred on mean vector. Bingham elliptical confidence cone is centred on maximum eigenvector on the symmetry plane connecting maximum and intermediate eigenvectors. (a) Density contours and statistics shown on both hemispheres using the split-hemisphere projection. This projection unhinges hemispheres at the E coordinate and inverts the upper hemisphere (see inset). The resulting contoured data reveal a distribution following the Bingham model. Therefore, the Fisher confidence cone and mean may not be representative. (b) Since there are data clusters in two modes, on upper and lower hemispheres, they are treated separately here.

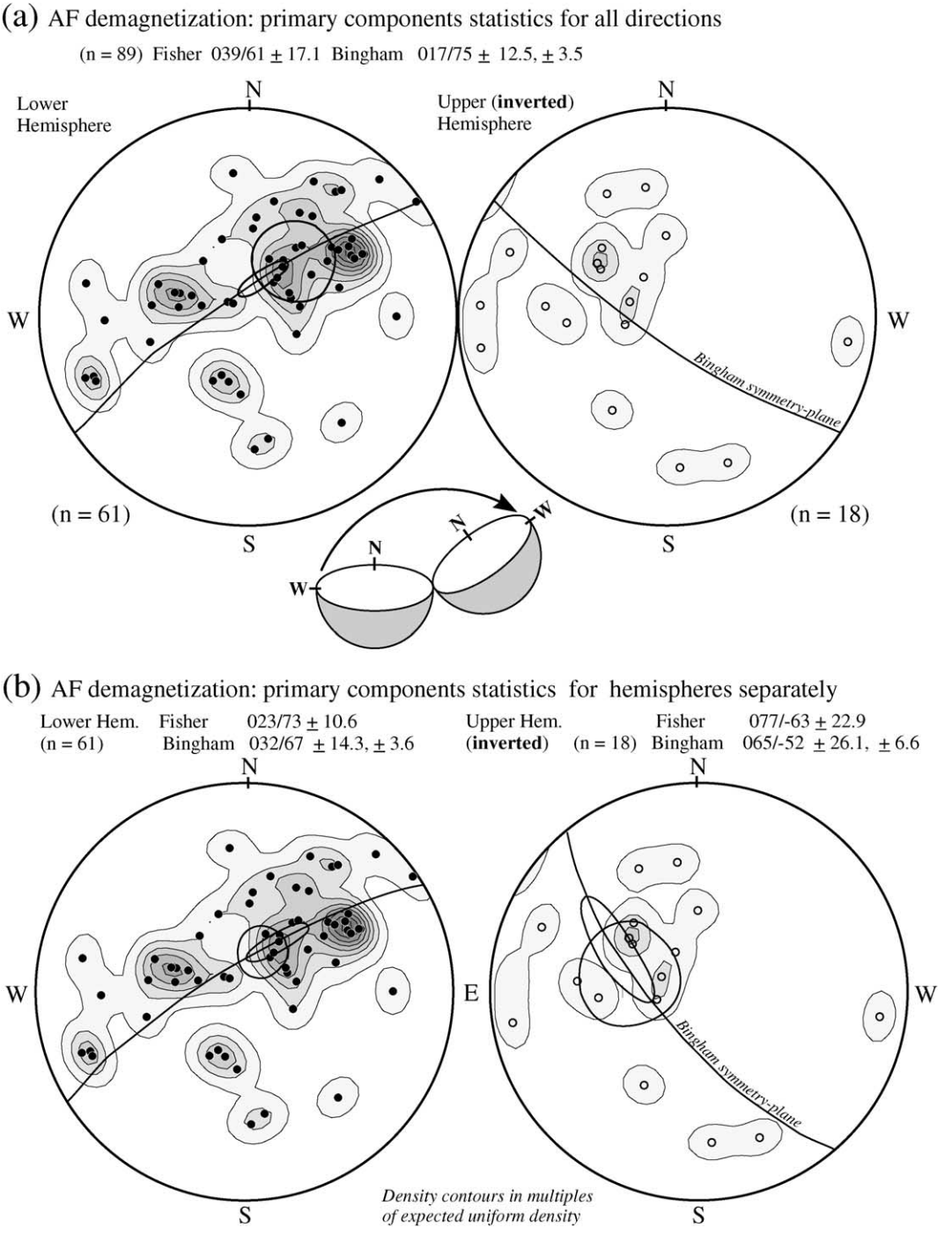


Fig. 5. New paleomagnetic characteristic vectors isolated by principal component analysis from alternating field demagnetization experiments using at least 20 demagnetization intervals per specimen. (a) Treating data on both hemispheres as a homogeneous group yields satisfactory statistics both by Bingham and Fisher models but the density distribution is more girdle like, favouring a Bingham population. (b) The same data, treating the modes on the upper and lower hemisphere separately.

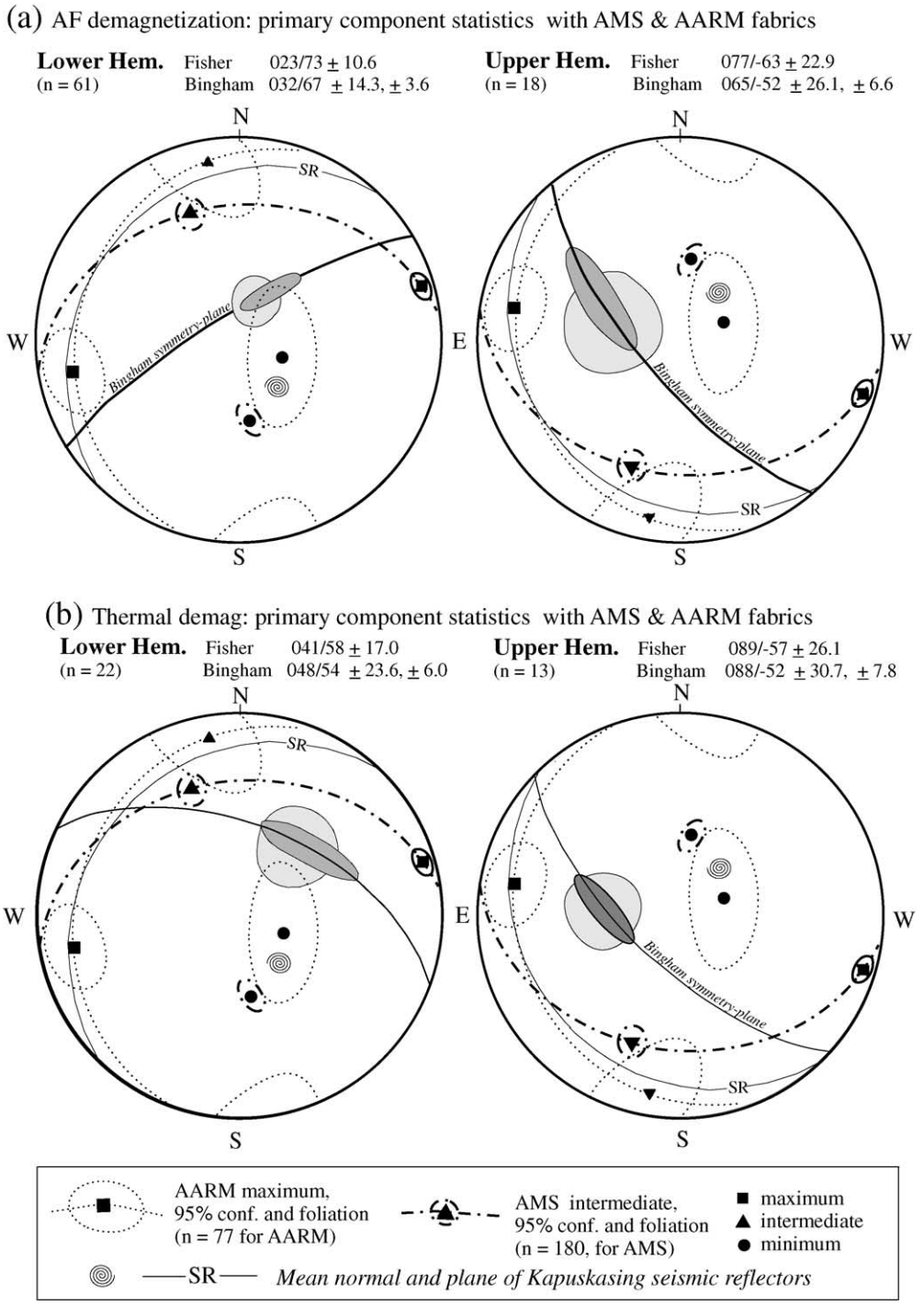


Fig. 6. Comparison of paleomagnetic vector statistics; vector mean and light-shaded α_{95} region for the Fisher model and dark-shaded elliptical confidence region centred on the maximum eigenvalue for the Bingham distribution. Comparison with magnetic fabric axes and magnetic foliations, seismic reflectors reveal the absence of tectonic influence on the paleomagnetic vectors.

cone is usually a good approximation for the Bingham model.

There are four possible causes of great-circle dispersion in this study, and their presence is readily detected using density plots on the split-stereogram projection (Figs. 4a and 5a). The first possibility is that magnetic anisotropy due to a metamorphic foliations may have deflected the paleofield. AMS was first developed to alert paleomagnetists to this possibility (e.g. Fuller, 1963, 1970). In metamorphic rocks, particularly high-grade granulites without platy minerals, a linear rather than planar fabric is more common. Nevertheless, a linear anisotropy may deflect paleofield toward the lineation, also causing an elliptical dispersion along a great circle.

Second, preexisting remanences may be dispersed by the effects of finite strain, where the finite strain ellipsoid has semi-axes $X \geq Y \geq Z$ (Borradaile, 1991, 1993, 1997; Werner and Borradaile, 1996; Cogné and Perroud, 1987; Hirt et al., 1986; Kodama, 1988; Lowrie et al., 1986). The concept is that, in some way, remanence vectors spin like passive line markers towards the maximum extension direction, as with conventional strain markers, e.g. acicular mineral grains. This simple model may be accepted where the remanence in a mineral grain is parallel to the grain's long axis, where the grains are large enough to rotate compatibly with a homogeneously straining rock mass and where stress does not disturb remanence preservation within the grain. This stretches credulity since we understand that the ChRM is a vector sum of spin moments distributed through magnetic domains in stress-sensitive grains and in subgrains that cannot strain homogeneously with the rock matrix at the macroscopic level. Nevertheless, some case studies from the field and laboratory confirm patterns of disturbed ChRMs that are compatible with the finite strain ellipsoid orientation in (op. cit. and Borradaile, 1997). Whether or not such simple ''homogeneously strained remanences'' actually owe their origin to that mechanism, they reveal that some component of the remanence, usually the hardest component (ChRM), is deflected towards the maximum extension X and away from the maximum shortening axis Z. However, following well-known concepts of progressive strain, line markers do not rotate from Z towards X by the shortest route, unless the finite strain ellipsoid is perfectly oblate ($X = Y$). Instead, the locus of the vectors is drawn towards the intermediate extension direction (Y) on its path towards X. Thus, according to small variations in strain and in initial orientations, the vectors may be scattered to lie on some arc someway between Z and X. In reality, the dispersion does not follow a great circle but, in practice, the pattern may be approximated by part of a great circle because the range of orientations is usually small.

Third, under stress, domain walls may be mobilized to accept a new remanence. Such stress-induced remanences are occasionally detected in field studies but they are also demonstrated in laboratory experiments (Borradaile, 1992a,b, 1994). The latter show that remagnetization adds vector components along a great circle connecting the field direction with the original remanence, regardless of the symmetry or orientation of the stress tensor. Experiments show that the softer remanence components are most easily replaced in multi-domain magnetite, although hard remanence components may be more amenable to stress demagnetization in hematite (Borradaile, 1992a,b, 1994; Borradaile and Mothersill, 1989).

Fourth, rocks from different sites within a region may acquire their remanences at different times for several reasons, including mineralization, changes in thermal or chemical conditions and localized mechanical events that may impose stress or strain remagnetization. The duration of these events may be similar to the intervals separating successive locations of the terrain during tectonic plate motion. Thus, paleomagnetic vectors from a single site may be smeared along the APWP. Vector components smeared in this way will mostly follow a great circle, which will be better defined, and more convincing, if reversals are detected.

Our samples of significant vectors isolated by thermal ($n = 35$) or by AF demagnetization ($n = 79$) are small in comparison to the sample size required in many other branches of science. For example, in structural geology, such small numbers of directions or axes would be considered inadequate to discriminate between unimodal and multimodal distributions, and between circular symmetry and partial-girdle distributions. However, in paleomagnetism, distributions are mostly more homogeneous, more concentrated and magnetic vectors have intrinsic constraints on the range of possible and reasonable orientation distributions, as discussed above. If the distribution is indeed homogeneous and coherent, the only reasonable pos-

sibility is that it lies somewhere in the range between a circular symmetry cluster and a girdle. In general, it should not be more complicated than a partial great-circle girdle with orthorhombic symmetry.

The mean direction of the vectors is obtained simply by vector summation, and this is adequate for the Fisher distribution. However, in the case of a Bingham distribution, the eigenvectors provide a more meaningful estimate of the principal concentrations. The minimum eigenvector is normal to the dispersal plane and the maximum eigenvector is parallel to the peak concentration. The Bingham 95% confidence limit is then centred about the maximum eigenvalue as an elliptical cone with its major axis (α_1) parallel to the dispersal plane. The aspect ratio (α_1/α_2) of the elliptical confidence cone reflects the degree to which the distribution is dispersed away from the maximum concentration (maximum eigenvector) along the symmetry plane. Unfortunately, the simpler calculation for the Fisher confidence cone always yields a value for α_{95}, even if the sample of vectors was not drawn from a Fisher population (Table 4).

In this study, ChRMs isolated by AF and thermal demagnetization appear to be dispersed along a great circle (Figs. 4a and 5a). Two modes appear in the lower hemisphere contours but it is difficult to determine their importance when so few vectors are included. Given the dispersed nature of the distribution and the expectations of a Bingham-population in such rocks, it would seem imprudent to use a Fisher model. This raises the issue of sampling and interpretation of orientation distributions. Paleomagnetic vectors from tectonized rocks may be reasonably expected to follow a Bingham distribution. The greater geometrical complexity of the elliptical distribution requires a larger sample of specimens. This may require a different sampling strategy from traditional paleomagnetic studies. For homogeneous Fisher distributions in sediments or volcanic rocks, multiple specimens from a few carefully selected individual sites provide the best statistical characterization. However, the dispersion of vectors in tectonized rocks, whether due to stress, strain or paleofield-refraction, is better represented by the widest possible spatial distribution of specimens within an approximately homogeneously strained domain. Thus, many sites should be sampled, preferably distributed according to some structurally determined sampling plan. This is implicit in earlier studies of the KSZ where sampling traverses are reported, in part corresponding to the seismic reflection profiles (e.g. Constanzo-Alvarez and Dunlop, 1988; Symons and Vandall, 1990). In our study, we used 40 sites determined by outcrop availability and structural position (Borradaile et al., 1999). Of course, pragmatism reduces the number of specimens per outcrop. For the 40 sites, we determined a total of 127 normal (down) ChRMs and 33 reversed (up) ChRMs which we term the A- and B-components, respectively. We also determined secondary vectors that are not further reported here as they are "soft" components corresponding fairly obvious to viscous remagnetization in Bruhnes' epoch geomagnetic fields.

Table 4
Selected Kapuskasing Characteristic (primary) remanences used for statistical study from 41 sites

	n specimens	Fisher *k*	Mean vector	α_{95}	Paleopole location	Max eigenvector	α_1, α_2	Paleopole location
A-component: lower hemisphere mode (normal directed remanences)								
Thermal	22	4.3	**041/58**	16.0	**009.2/59.1**	048/54	23.6, 6.0	009.8/52.2
A.F.	61	3.2	**023/73**	10.6	**320.6/72.8**	032/67	14.3, 3.6	350.4/69.0
B-component: upper hemisphere mode (reverse directed remanences)								
Thermal	11[a]	3.5	**088/ − 51**	23	**210.6/22.0**	088/ − 51	33.1, 8.2	210.7/21.7
A.F.	18	3.0	**077/ − 63**	16	**227.2/24.3**	065/ − 52	26.1, 6.6	226.3/9.3

Bold letters key to locations on APWP map, e.g. paleopole locations defined by **T**hermal demagnetization and **B**ingham statistics are denoted **TB**. We used at least 12 steps in each thermal demagnetization experiment and at least 20 in each AF demagnetization test. Remanence directions were determined by principal component analysis (PCA), Kirschvink (1980). Site mean latitude and longitude are 48°N and 277°W. Vector mean involves no structural correction.

[a] Two outliers deleted (see Fig. 4b).

3. Our interpretation

Thermal and AF demagnetization yields different distributions of primary A-component vectors, and in each case, the orientation distributions of A-components have either multiples modes or some structured distribution that does not conform to the ideal symmetrically circular cluster of points on the sphere. In these circumstances, replotting the orientations, centering them on their mean direction (Fisher et al., 1987) or on some important structural axis may be easily accomplished in computer software such as we use (Spheristat by Pangaea Scientific). However, our data are conveniently distributed with concentrations of paleomagnetic vectors that are steep, either up or down, so that we may retain the geographic frame of reference and simply use Fisher et al.'s split hemisphere technique to flip over the usual upper hemisphere. Of course, this will not be possible in general; in other studies, especially where concentrations have low inclinations, data should be centred before adopting the split-hemisphere presentation.

The ChRMs isolated by thermal demagnetization (Tables 1 and 2) are illustrated in Fig. 4. The interpretation of the density distribution of vectors on a Fisher-split net (Fig. 4a) reveals that the reversed (right net) and normal remanences (left net), although they are not an anti-parallel pair, could lie on the same great circle. This is difficult to recognize from conventional stereographic projections using a single net. The Fisher mean, with its large confidence cone centred on 055/25, is meaningless, and the presence of two modes (upper and lower hemispheres) would have alerted us to this anyway. However, the Bingham statistical model might satisfy the distribution of all data, both normal and reversed, with a tight confidence ellipse. The separate treatment of the two modes (Normal and Reversed polarities) is shown in Fig. 4b. It was possible in this study to separate the data according to its distribution on the upper and lower hemispheres. In the general case where modes are not steeply inclined, it would be necessary to centre the data and then split the sphere about some plane that was not originally horizontal.

The lower hemisphere projection (left) appears to be best explained by a Bingham model because the data disperse along a great circle with the confidence ellipse symmetrically located in the density distribution. However, the reversed directions shown in the inverted upper hemisphere (right) are more agreeably modelled by a Fisher distribution, there being no indication of dispersion along a plane. Therefore, the Bingham model is preferred for the normal remanences but the Fisher model is satisfactory for the reversed vectors. This makes some difference to which mean direction we accept, and as we shall show, to the position of the KSZ on the APWP.

The results for characteristic vectors isolated by AF demagnetization are shown in Fig. 5. In all cases, confidence cones are smaller due to the larger sample size and homogeneous nature of the density distribution of vectors. In Fig. 5a, all data including the normal and reversed modes are treated as one sample. The Fisher model produces a sensibly located mean vector with a modest α_{95}. However, the distribution clearly follows a great-circle girdle that suggests that the population is Bingham in nature. Bingham statistics define a symmetry plane that passes symmetrically not only through the lower hemisphere data (left net) but also through the reversed mode on the upper hemisphere (right net). The elliptical confidence cone is agreeably small. When we split the data according to the two main modes, one normal the other reversed as in Fig. 5b, the Bingham model is very satisfactory for the lower hemisphere (normal vectors), but the Fisher model is preferred for the upper hemisphere (reversed vectors).

In conclusion, the normal remanences seem to follow a Bingham model due to dispersion along a north-dipping plane. However, the plane strikes NW–SE for thermally isolated vectors and NE–SW for vectors isolated by AF demagnetization. These are the "A" or older Archean signals of other authors. In contrast, the reversed remanences, the younger, late Archean or Proterozoic "B" chemical overprint of other researchers appear to be drawn from a Fisher population, there is no tendency for the vectors to smear along a great circle.

For the moment, we shall not debate the merits of thermal versus AF demagnetization, which is a well-discussed aspect of rock magnetism that must be decided with a knowledge of the history of magnetic mineralogy (Butler, 1992; Dunlop and Özdemir, 1997). It does not attempt to undo the natural magnetization process but rather sets the artificial goal of separating distinct vector components according to

their relative age. Different laboratory techniques including the familiar thermal and alternating field demagnetization methods, as well as less popular chemical and low temperature methods, all may isolate differently oriented vector components. In this study, there are differences between mean directions isolated by AF and thermal demagnetization, particularly for the A-component (normal remanences, Figs. 4b and 5b). Furthermore, we are concerned that the differences due to different statistical treatments of the same data are almost as great as the differences due to demagnetization techniques. This may be observed by comparing the Fisher and Bingham confidence regions on the same stereograms (Figs. 4b and 5b).

This is summarized in Fig. 6 where the relative confidence cones about the mean vector are shown for each treatment in different shades of grey; lighter gray for Fisher confidence region and darker gray for Bingham region. For reference, AMS and AARM foliations and the mean orientation of seismic reflectors are shown. We believe that the magnetic fabrics are an expression of the metamorphic foliation, which is otherwise difficult to observe in granoblastic rocks, in the field, due to the absence of platy minerals. The similar orientations of the three planar structures are encouraging, not withstanding the inaccuracies in determining the true dip and strike of the mean seismic reflection surfaces from two-dimensional profiles and depths inferred from two-way travel times.

There appears to be no simple angular relationship between the normal polarity (“A”) hard remanences on the lower hemispheres (Fig. 6a and b, left sides). That is, Fisher or Bingham confidence cones do not correspond to any principal anisotropy axis (AARM, AMS or seismic reflection plane). Moreover, the Bingham dispersal plane is unrelated to the foliations or principal axes of either magnetic anisotropy (AMS or AARM). This is encouraging for paleomagnetism as one may deduce that there is little anisotropy deflection of remanence during its acquisition. However, the distributions are different for vectors isolated by thermal and AF techniques. Furthermore, one could deduce that there is little possibility of post-acquisition deflection of remanence by tectonic strain (or stress). Similarly, the B-components (reverse remanences, right stereograms, Fig. 6a and b) appear uninfluenced by anisotropy, but the Bingham mean directions differ for thermal and AF-treatments.

The most compelling manner in which to evaluate paleomagnetic results is to inspect the paleopole positions on a world map and compare them with the available apparent polar wander paths. Two cautionary notes are in order. First, any map projection will compromise comparisons. Where paleopoles lie on one hemisphere, spherical projections of the globe suffice. Here, the paleopoles are so widely distributed that a plane map projection is preferred, with the disadvantage that there is enormous distortion of longitudinal differences at high latitudes. Furthermore, APWPs for the late Archean and early Proterozoic for the Superior Province (Fig. 7) are extremely tentative by comparison with APWPs for Phanerozoic time, and there have been some considerable revisions over the last 30 years in the light of refined geochronology (Fig. 8). The difficulty of performing stability tests (even reversal-tests), the long metamorphic and thermal history and the difficulty of applying tilt where the tilt-axis is poorly known, all constrain APWPs very loosely by comparison with those of Phanerozoic time. Nevertheless, the APWPs provide a useful frame of reference based on a consensus of published results. Fortunately, almost all studies have sensibly avoided and excluded remanences that may have been affected by penetrative deformation, the finite strain of structural geology. Finally, it is only since the mid-1980s that reliable high-precision U–Pb geochronology has been able to determine the age of magnetization-events. Even so, considerable are apparent; largely before high precision U–Pb geochronology, an Archean APWP was documented that traversed from the south Atlantic through Africa and Asia between 2700 and $\sim$2000 Ma (Irving, 1979; Dunlop, 1983, 1984a,b, 1985, Fig. 8a). Refinements in light of more precise geochronology and new paleomagnetic data revised the Archean APWP for 2700–2400 Ma to curl over Siberia and Scandinavia (Symons et al., 1994; Vandall and Symons, 1990, Fig. 8a). These paths are shown because they are still valid loci for contemporary paleopole determinations for post-metamorphic cooling remanences ($\geq$2700 Ma) of highly deformed metamorphic rocks in northern Ontario.

More reliable paleopole estimates come from late-orogenic and cratonic dikes and other minor intrusions, for which tilt corrections are negligible or unnecessary. Reliable, high precision geochronology is now available for some of these rocks. Conse-

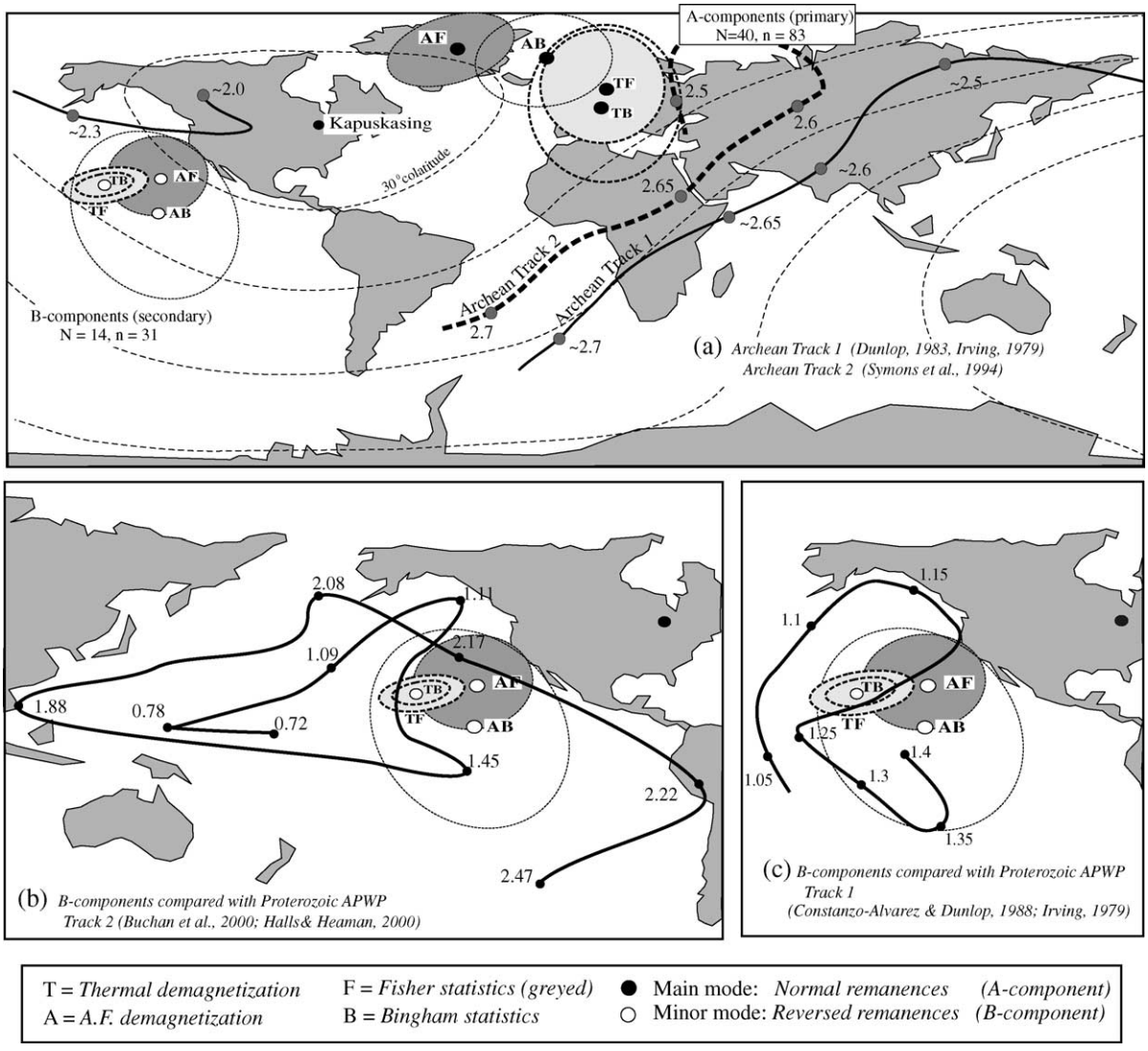


Fig. 7. (a) Paleopole locations from vectors isolated by AF (A) and thermal (T) demagnetization using statistics based on the Fisher and Bingham models (F and B, respectively). These are separated simply into positions corresponding to one normal and one reversed mode in each orientation distribution of vectors. Note that paleopoles determined by thermal demagnetization tend to lie further from the site. For the A-components, paleopoles locations are quite different, depending on the demagnetization technique chosen. Tentative apparent polar wander paths for the Superior Province, from some published sources are shown for available Archean and Proterozoic rocks in the Superior Province. Dashed lines show equal colatitude with respect to the site. (b) Paleopoles from B-components compared with a recently determined high precision APWP mostly defined by high precision U–Pb geochronology. Even so, precision is variable at different parts of the path and the reader is advised to consult literature where necessary. (c) An older version of the Proterozoic APWP compared with our B-components.

quently, the Archean APWP and the early Proterozoic APWP have been revised as shown in Fig. 7b (Buchan et al., 1993, 1996, 2000, 2001; Halls and Heaman, 2000, Fig. 8b). The Proterozoic APWP is fairly well constrained between 1400 and 900 Ma. However, data are still sparse and even the latitudinal component of apparent wander may be so fast so that the actual locus is uncertain (Figs. 7b and 8b). Until recently, the Proterozoic APWP used in most tectonic evaluations used the compilation in Fig. 7c (Bates and Halls,

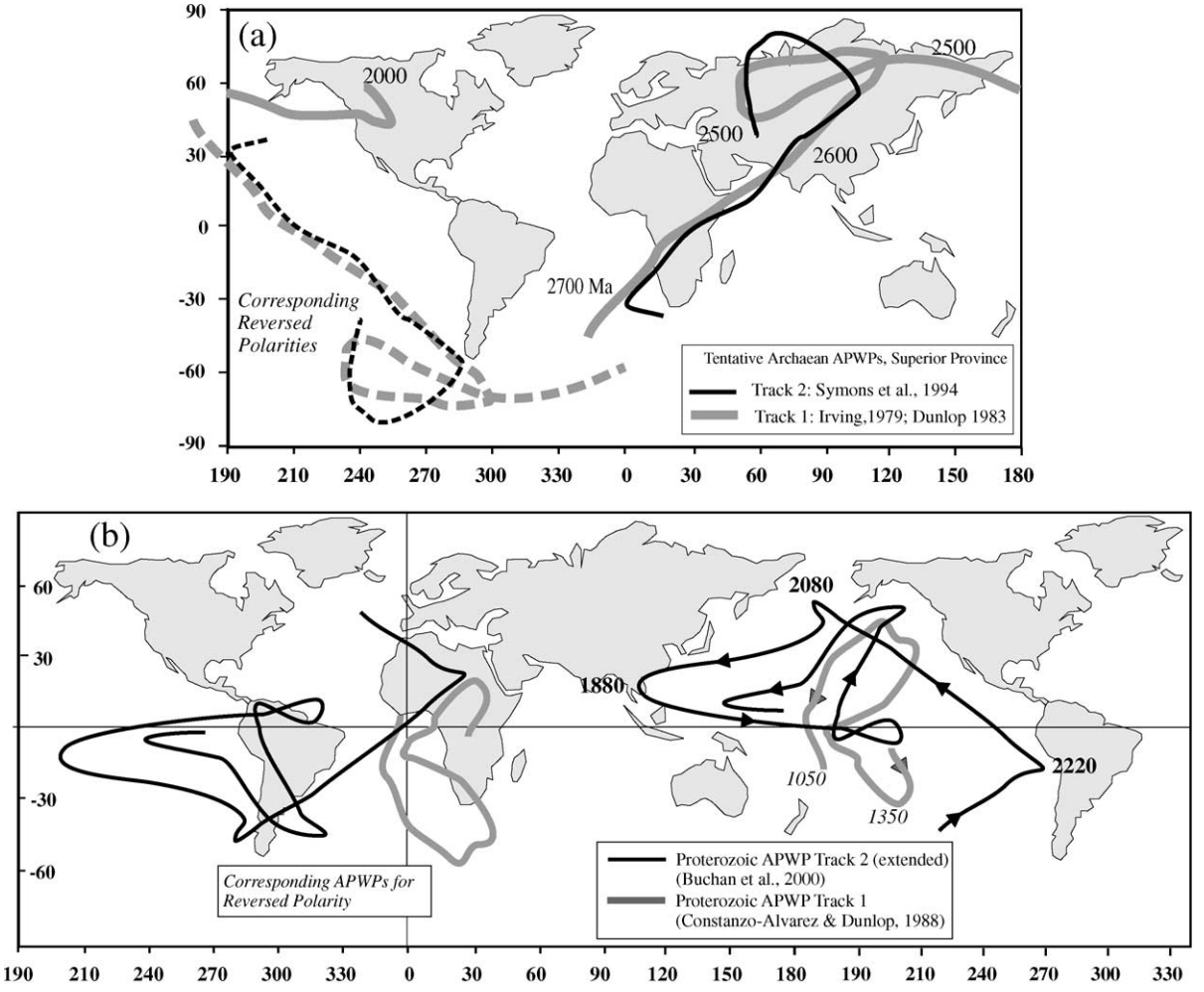


Fig. 8. Comparison of the available published APWPs and their antipodal traces on which reversal paleopoles may be located. (a) For paleopoles determined from the Superior Province Archean rocks. The geochronological data available at that time render these paths much less certain than those constructed in the 1990s, with the benefit of U–Pb high precision geochronology. Nevertheless, these paths still compare well with many contemporary paleopole determinations from strained rocks in northern Ontario. (b) Recently determined Proterozoic APWP (Buchan et al., 2000) for which geochronology is precise and earlier version of the path (Constanzo-Alvarez and Dunlop, 1988).

1991a,b; Constanzo-Alvarez and Dunlop, 1988; Halls, 1986, 1991; Halls and Shaw, 1988; Halls et al., 1994; Pesonen and Halls, 1979). Given the precision of some paleopole locations, this does not always differ too greatly from the more selective database using high-precision geochronology shown in Figs. 7b and 8b.

Consider the B-components (reversed polarities) attributed by others to remagnetization during Proterozoic uplift with higher coercivities perhaps suggesting a CRM origin. Their mean paleopole positions lie close to part of the Proterozoic APWP segment in Fig. 7b, which corresponds to 1400–1200 Ma. A younger age of ~1100 Ma was preferred using the previously suggested APWP (Fig. 7c). Their proximity to the actual is surprisingly agreeable, despite the large areas encompassed by the 95% confidence regions. The paleopoles' confidence regions for the two experimental treatments (AF and thermal) and two statistical treatments (Fisher and Bingham) show considerable overlap despite sub-

stantial differences in their mean location. One might argue that the thermal treatments favour positions that are closer to the recently defined path (Fig. 7b) and at slightly younger locations by referring to the older APWP (Fig. 7c).

Next, consider the A-components which have normal polarity. These generally have higher unblocking temperatures and lower coercivities than the B-components and, in previous studies, were attributed to the early acquisition as the Kapuskasing gneisses unroofed during the Late Archean, through a critical isotherm that is related to the experimental blocking temperature. Clearly, the mean paleopole locations and their 95% confidence regions do not lie on any part of the published APWPs (Fig. 7a and b). Tectonic and paleomagnetic literature on the KSZ has offered varying tilt corrections for these early vectors, e.g. 15–30°NW by Constanzo-Alvarez and Dunlop (1988) and ~10°NW by Symons and Vandall (1990). If a tilt-correction is required for the A-component, we find little structural information to discriminate between those suggestions; neither value significantly affects paleopole locations. Initially, a tilt-correction seemed mandatory. The upthrust lower crustal gneisses are tilted on a NW-dipping ramp structure (Fig. 1 and Atekwana et al., 1994). However, the magnitude of the ramp was not at first understood, and since the magnetic fabrics and seismic reflectors are subparallel, we suggest that all the microfabrics, visible, AMS and AARM, extend to similar depths (Fig. 1). This makes it most unlikely that the entire KSZ ramp was steepened to its present orientation by tilting that post-dated metamorphism and the blocking-in of the A-components. Based on the common maximum blocking temperatures for the early remanence (>400 °C, Table 1), a conservative geothermal gradient of 30 °C/km predicts blocking temperature isotherms at a depth of ~13 km. The orientations of the microfabrics persist to these depths, even now. Prior to denudation, their orientation would have been retained even deeper into the crust. Therefore, we suggest that a tilt correction may be unnecessary. Even if a tilt-correction was applied, it would deflect the A-component vectors so that their paleopoles would lie still further westward, not eastward, and the disagreement with the Archean APWP would be greater, not less (Fig. 7a). At present, we can offer no explanation for the discrepancy between our early paleopole locations situated in the North Atlantic region and the closest part of the APWP (2600–2500 Ma) in Siberia (note the large longitudinal dilation of locations at high latitudes in the map projections). Most seriously, mean paleopoles determined by thermal demagnetization differ from those fixed by AF demagnetization, significantly in the case of the A-components (Fig. 7a).

4. Conclusions

First, the seismic profile shows that the thrust ramp and its associated magnetic–petrofabric foliation has constant inclination to depths that probably exceeded the blocking-in isotherm depths for the A-component. This means that the uplifted granulites were magnetized in their present orientation and a tilt-correction for their A-component is probably unnecessary. Moreover, correction for any WNW tilt on the ramp would increase the discrepancies between our A-component paleopole locations and the tentative Archean APWP (Fig. 7a).

Second, despite well-defined AMS and AARM fabrics, there is no obvious anisotropy-deflection or tectonic-deflection of either the A-component or B-component remanences. The A-component TRM and B-component chemical overprint may be regarded as reliable indicators of paleopole locations.

Third, although we used the same specimens for different treatments, thermal demagnetization produces differences from AF demagnetization that gives some concern. The merits and disadvantages of those treatments and others, such as low-temperature demagnetization and chemical demagnetization, are discussed in the literature (Butler, 1992; Dunlop and Özdemir, 1997; McElhinny and McFadden, 2000; Van der Voo, 1993). The choice is even more difficult in metamorphic rocks where magnetite grain-size is not necessarily a strong clue to the age of the mineral or its remanence. For example, hard remanences may be carried by fine grains that grew late in the metamorphic history and the effects of stress may "magnetically harden" grains. On its own, differential stress may remagnetize grains. Thus, a coercive or refractory remanence may not necessarily be ancient where metamorphic and tectonic processes have been active. The differences

between paleopole locations are small for the B-component (Fig. 7b and c), but the overlap of the α_{95} confidence regions for the A-component is unsatisfactory (Fig. 7a). However, the sense of discrepancies between paleopole locations defined by thermal demagnetization from those defined by AF demagnetization is consistent. Thermally defined paleopoles lie further from the site than those defined by AF demagnetization; they correspond to shallower inclinations. Inspection of the paleopoles in relation to the lines of co-latitude about the Kapuskasing site clarifies this (Fig. 7a).

Finally, using the same data, the determined paleopole locations and confidence regions differ according to statistical treatment. Since statistical treatments are non-destructive experiments, it seems prudent to inspect all paleomagnetic data from strained rocks, routinely, for any tendency to form a partial girdle that would suggest the sample was drawn from a Bingham population. For the B-components, the mean paleopole locations lie in the NE Pacific and their α_{95} confidence regions coincide when two outliers are excluded (Fig. 7b and c). They appear to correspond to the Proterozoic APWP at magnetization ages of 1400–1200 Ma (Fig. 7b). Earlier published opinions ranged from 1050 (cf. Fig. 7c) to 1860 Ma (Table 2). However, whichever statistical procedure is applied to the A-component remanences, their mean location and confidence regions are influenced significantly by the choice of demagnetization treatment (Fig. 7a). A-component paleopole locations lie far from the tentative and poorly constrained APWP, for which the best matching age is $\sim$ 2500 Ma.

Acknowledgements

We are especially grateful to David Symons' suggestions and reading of an early version of the manuscript. Ken Buchan and Rob Van der Voo provided helpful, critical reviews. This work was funded by NSERC operating and equipment grants to Graham Borradaile. Courtney Cameron (née Goulder) helped in field sampling. Anne Hammond produced excellent drill core from oriented field-specimens, as well as mineral separations for Curie balance tests and petrographic thin sections.

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